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A Voyage into the Visualization of Athletic Performances: A Review

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ABSTRACT

Every athlete's path includes visualization as a tool for improving performance and developing plans to achieve their goals. Visualization assists individuals in visualizing how they wish to perform in their chosen sporting event. Every athlete believes and understands that visualization is what they require to achieve their goals. In this review paper, researchers examine both quantitative and qualitative research methods to describe the power of visualization; why it is important for every athlete; visualization methods used by athletes; the effects of visualization on athletic performance; and visualization techniques in sports. A literature review was conducted on several published peer-reviewed journal articles, but only articles from 2015–2022 and those written in the English language were considered for inclusion in this study. Researchers made use of search engine like Google Scholar, Microsoft Academic, ProQuest, PLoS One, Cochrane Library, and Microsoft Academic databases for information. Several articles were related to the subject, but only a few text articles were used for this review. In conclusion, the reviewed articles demonstrated that what coaches, trainers, physical educators, and sports teachers need to do is provide more assistance to help our athletes build more effectiveness in visualization to improve their sporting performance, which will serve as a fantastic tool that they may utilize to establish the mentality needed to attain long-term and short-term goals and teach younger athletes visualization, it's important to do it in a way that's easily accessible and fun for them.

INTRODUCTION

For good reason, visualization has become a term in sports psychology and other areas of performance. Visualization is being used and credited by an increasing number of athletes across all sports to improve outcomes and achievement. Sports have never been more competitive than they are today, with victory margins narrowing as we reach the pinnacle of human speed, strength, and agility. Anything that can provide you an advantage should be pursued and incorporated into your training regimen, and the importance of psychology cannot be overstated.

A clear mind works best for intuitive, free-spirited, and experienced athletes. They have a lot of confidence and belief in their abilities because of their many years of accomplishment. You develop a peaceful mentality by thinking on something other than your sport. Talking to others, thinking about someone or something that makes you happy, and listening to music in your brain are all ways to keep your mind clear and avoid it from interfering with your body's performance.

It is critical to visualize your accomplishment in sports activity. Visualization will assist you in planning your performance in the sporting event. Every athlete takes several measures to instill and master. To establish which attitude will work best for you, you must first experiment. Second, you must make a commitment to cultivating a good mindset. Third, you must focus on building your intended goal during practice and competition. Finally, you must train and compete enough to deeply embed your ideal attitude, so that when you enter the most important competition of your life, it simply turns on and allows you

to perform at your best. It has a huge impact if an athlete visualizes before and after a competition. People from all areas of life have publicly introduced visualization. They were sports psychologists, coaches, and motivational speakers in 1980. (Competitive Edge, 2021).

Visualization in sport, according to Quinn (2021), is a training approach that is part of the greater field of sports psychology. It is the practice of visualizing, guided meditation, and setting an intention for what you want to happen or experience. Visualization is a person's point of view in which they rehearse the intended outcome, similar to reading a screenplay that contains emotion (The After Bar, 2022). There are two sorts of visualization: outcome visualization (visualizing the outcome) and process visualization (using the other senses of the body to process the desired conclusion) (Moe, 2021). Furthermore, visualizing is a powerful strategy that anyone in a competitive context can apply. In reality, the world's most successful athletes imagine their success before a tournament. In contrast to quiet meditation, which entails letting go and not intentionally guiding your thoughts, visualization entails consciously forming mental images (Erikson, 2019).

Visualization, often known as visual imagery, is the technique of creating mental representations of a desired outcome in your mind. Mental imagery is a sort of psychological intervention that can help with depression, anxiety, and other problems (Skottnik & Linden, 2019). These images are created by mentally rehearsing the game in your head while using your five senses (Rollins, 2016). Michael Phelps, the most successful Olympian in

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history, has employed imagery brilliantly (Predoiu et al. 2020). Visualization can also be considered as a type of training and warm-up that helps athletes improve their performance (Cohn, 2022).

According to Dr. Blaslatto of the University of Chicago, who led the groundbreaking visualization research. He assessed players' progress in free throw shooting in three groups. One group did not practice for 30 days, while another spent 30 minutes per day on free throws and the third imagined successful free shots. The group that did not practice made no progress, the group that conducted physical practice made 24 percent more progress, and the group that imagined made 23 percent more progress (Razmus, 2020). Furthermore, many professional athletes use visualization techniques as part of their preparation and competition. Many athletes have used these tactics to build a competitive edge as a multi-sensory experience (Quinn, 2021). Indeed, professional or elite athletes have mastered and applied visualization or imagery in every game (Cohn, 2022). Phelps' final in the 200m butterfly at the 2008 summer Olympic Games, according to Predoiu et al. (2020), was arguably the strongest example of the power of mental imagery. Visualization can help athletes enhance their motor skills, self-confidence, attention, concentration, and anxiety, as well as their pain management, physical motivation, and performance. It also makes athletes calmer and more adaptable to tough situations. However, the process entails far more than simply visualization (Quinn, 2021).

With this scenario, the researchers aimed to reveal how every sports enthusiast can support the effective of visualization when used as method to enhance sporting performance.

Power of Visualization

The brain is an asset in both sports and life. Individuals with a performance mindset have developed methods to harness this drive to propel them forward. Mental imagery and visualization skills have been researched and proved to be valuable advantages for athletes preparing for competitions in sports as diverse as boxing, basketball, and volleyball (Weiland, 2016). In fact, guided imagery can be used to help people visualize themselves in a safe or relaxing situation, with the goal of reconnecting with this experience later when it will be useful. It could also be seen as a strategy to deflect attention away from bad emotions (Ekeocha, 2015). According to Mallette (2020), the traditional way of practice is that repetition of one thing leads to skill development. However, just as our bodies are complex, so are our minds. Visualization occurs when the mind imagines something, and the body responds. Visualization has been used by successful people.

Indeed, visualization has tremendous potential in creating an environment conducive to athletes achieving their life goals (Canfield, 2022). The mind is quite powerful. When used effectively, visualizations may be a powerful tool for success. Successful people are aware of the importance

of visualizing. They are more likely to succeed in their ideas before interacting (Mayberry, 2015). Athletes, entertainers, surgeons, doctors, police officers, and politicians all believe in the power of the mind. "If my mind can imagine it and my heart can believe it, then I can achieve it," said Muhammed Ali. Affirmation and visualization are powerful tools that celebrities such as Arnold Schwarzenegger, Oprah Winfrey, and Jim Carey believe in. Immunity, stress, healing, and pain management have all been proved in scientific research to have positive benefits on health (Globokar, 2020). It improves several elements of life, such as athletic ability, cognitive function, self-esteem, and goal attainment (Smith, 2018).

According to Emily Cook, a three-time Olympian, "the word visualization, for me, does not take in all the senses." You must smell it. You must hear it. Everything has to be felt. "She excels at freestyle skiing and values visualization, even taking it to the highest level (Restoic, 2020). Michael Phelps, the most decorated athlete in history and an American swimmer, is also proof that visualizing works. He couldn't concentrate at that age. Then, his coach, Bob Bowman, introduced Michael to visualization and advised him to imagine his best performance in his head. Michael Phelps visualizes finishing a triumphant race every night, and it is quite effective in reality (Athlete Authority, 2022). In any situation, visualization reigns supreme. It will condition your brain and assist you in responding. It will psychologically prepare you for a specific situation (Rollins, 2022). According to Dr. Michael Gavais, a sports psychologist, the goal of visualization is not to anticipate the future, but rather to prepare for unexpected actions from the opponent ahead of time (Feloni, 2020).

As a result, visualization increases the likelihood of success. The late great was recognized by his persistent work ethic, as proven by Kobe Bryant. He was well-known for being drenched in sweat before 5 a.m. every day and for having drilled so thoroughly that he held himself to shooting 800 shots before tipoff every night. Many people are unaware that Kobe worked outside of the gym, the court, or the editing room. Some of his most important work was done alone in a quiet room with his eyes closed, picturing the game to be played, a tendency shared by Michael Jordan. Even before he was given the honor, Jordan remarked that he had always imagined himself taking the game's final shot. He saw himself in the biggest moments, making the biggest shots, and he was ready when the time came (Felber, 2020).

The reasoning behind this is that there is strength when an athlete can envision the right sport's competitive as well as the flow and end of the game while picturing. Athletes' defenses and strategies are already in their heads. A good visualization does more than just give a visual interpretation of data; it also improves comprehension, communication, and decision-making (Stewart, 2019).

Importance, Benefits, and techniques of Visualization for every athlete

The importance and benefits of visualization have been proved to aid in increasing wellbeing and overall functioning, but it may also be used to non-health situations. It simply allows athletes and coaches to visualize themselves achieving any difficult activity that may appear insurmountable or life-threatening. It has been shown, for example, to be particularly beneficial in enhancing athletic abilities.

Visualization techniques are vital for all athletes and their coaches because they can help enhance motor abilities, muscle strength, self-confidence, attention span, anxiety, and stress. Athletes can improve their pain management, endurance, performance motivation, and physical performance by using imagery (Predoiu et al., 2020). According to Predoiu (2016) highlighted some benefits and techniques of visualization that can be useful among athletes:

- i. The focus of attention increases, athletes being required to reach their full potential,
- ii. Athletes learn to be more positive (mental practice is characterized by positive and Positive thoughts are eliminated because of successful representations,
- iii. Athletes better understand the importance of goal setting,
- iv. Self-control ability is improved, tension and fear are reduced, while self-confidence grows (Xiong, 2012),
- v. The athlete prepares the optimal programme of technical and tactical execution (Mental practice helps to improve temporal and spatial organization) motor action characteristics,
- vi. The athlete's thinking becomes more disciplined, organized, and planned, raising awareness of the motor skills performed,
- vii. The time needed to acquire different technical and tactical elements is reduced, and the level of preparedness for stressful situations (a characteristic of competitive activity) is increased.

Sowinski (2018) also mentions the usage of mental imagery as a visualization tool. This approach is usually utilized by soccer players to memorize the game while watching or studying it. This is a strategy for making your desired future a reality (Moe, 2021).

Furthermore, there are other methods for visualizing; you must be explicit with your goal, use all five senses, do it in real-time with no skipping or fast-forwarding, and practice often to make it a habit (Restoic, 2020). A visualization approach can assist the athlete in controlling their fury. They may see themselves maintaining calm while an opponent attempts to start a fight (dot FIT, 2022). Meanwhile, guided visualization is a stress-reduction approach. It's a soothing approach that involves picturing pleasant, quiet places, such as a beautiful beach or a peaceful meadow. This technique is sometimes referred to as guided meditation or visualization (Nunez, 2020). Most people begin by relaxing in a comfortable and calm environment, such as a comfortable chair, bed, or even a hot tub (Ekeocha, 2015).

According to Parnabas et al. (2015), athletes at higher

levels were more likely to employ mental visualization techniques in their sport, meaning that less experienced athletes may not benefit as much as their more experienced counterparts. It is normal for an athlete or player to have negative thoughts, anxiety, and even panic before or during a game. A coach, on the other hand, should always prioritize mental preparation for his or her athletes. Before a game, visualization plays a crucial role in a team. A coach cannot ask a player to use or exercise visualization immediately and must adequately explain its value to the athlete. If the athlete is ready, the coach can determine whether or not to incorporate it into the game (Peterson, 2017).

Collins (2017) demonstrated that certain scenarios are frequently outside the athletes' control, and the stress they generate can be worsened by something more sport-specific, such as being a pinch hitter with the bases loaded in the ninth inning. So, how do athletes maintain their calm and confidence in these uncertain situations? Some professional athletes may argue that they control what they can and let the rest happen spontaneously; others, on the other hand, prepare for them through mental training, particularly imagery. Every day, these athletes will see and construct events in their minds' eyes, preparing them for every eventuality. However, learning to control your own thoughts is a talent in and of itself. Visualization is a method for dealing with bad emotions (Peterson, 2017). When an athlete visualizes a situation, he or she does so in two ways. Visualization of unexpected results and mental practice of a flawless workout (Predoiu et al. 2020). Athletes often visualize how their team prepares for competition, and visualization can assist an athlete remember a game and the overall sensation of a successful outcome (Peterson, 2017). Having the opportunity to recreate a game will allow athletes to develop themselves (Tilkes, 2020). Dr. Gevais stressed that practicing visualization ahead of time is more helpful than practicing on the same day, which is unsuccessful (Feloni, 2017). We cannot dispute the link between the body and the mind. When you understand how to improve your thinking, you have complete power over your performance (Mallette, 2020). As a result, visualizing an event ahead of time can help you perform better mentally and physically (Cumming, 2016).

Methods of visualizations used by Athletes

Athletes, on the other hand, have employed a variety of strategies to picture their games and prepare their brains for impending competitions. To achieve the best results, visualization approaches should encompass the five major senses (touch, hearing, sight, smell, and taste) as well as critical features such as viewpoint, mood, surrounds, task, and timing. Mental rehearsal (or visualization) is effective because the subconscious processes the experience as if it were real (by firing the neurons responsible for skill acquisition), making the person calmer and more adaptable to stressful situations, and it can speed up the learning process in athletes and others. One hour

of mental training per day in 6-10 sequences provides a distinct advantage that cannot be obtained by any other way. According to Predoiu et al. (2020), athletes employ guided imagery techniques and practice to create mental maps that help them perform effectively. Visualization techniques that work for athletes can benefit all athletes, regardless of their goals or performance level.

Here are five (5) tried-and-true strategies for visualizing like a pro: tips to guide you along your visualization route.

1. *Obtain Clear on Your Objectives:* You must be clear about what you want to accomplish. It could be a specific technique you're working on or a specific incident you want to nail down. When you are specific and detailed, visualization works best. It should be as realistic as feasible.

2. *Utilize More Than Just Images:* All your senses must be engaged. Visualization encompasses more than just images. The experience must be visible, felt, smelled, heard, touched, and even tasted. If necessary, use physical actions to make it feel more real.

3. *Execute It in Real-Time:* Do not attempt to hasten the procedure. A 200-meter race should take more than 20 seconds to complete. Similarly, if you're ticking down the seconds until the game-winning jump shot, the visuals should be in sync. The visualization should last if the actual race or event. Don't scrimp on quality. If you want to harness the power of imagination, talk to a mental game coach about incorporating this important ability into your training.

4. *Practice, practice, and more practice:* Visualization is a skill that takes time to develop. Do it every day, make it a habit, and strive towards near reality. Mental rehearsal for athletes is a skill that improves with practice. Every day, practice your visualization or picture.

5. *Make it a routine to Visualize the outcome:* Visualization can help you perform much better. It can help you with both the physical and mental skills you require by mentally rehearse your performance in your head. So, whenever you have a few quiet minutes, make it a habit to visualize your success. If you need to take a pause and see the event as how it will unfold. If your mental images turn negative, stop the mental tape, rewind and restart then visualize again see the performance you want to see. to heal from an injury or simply need a physical break, employ visualization to keep your practice going. Psychologists have discovered that significant mental practice is nearly as effective as the real thing, making it one of the best tools for honing your talents.

CONCLUSIONS

According to the review papers, visualizing has a significant impact on every athlete who want to succeed in their various sports. It made a substantial contribution to the theoretical understanding of some mental processes explored in athletes' sports engagement during competition. It also demonstrates that the most effective visualization techniques should result in an incredibly vivid sports experience in which the athlete has entire

control over achieving achievement. Visualization has shown to be a useful tool for general technique practice. In conclusion, the reviewed articles demonstrated that what coaches, trainers, physical educators, and sports teachers need to do is provide more assistance to help our athletes build more effectiveness in visualization to improve their sporting performance, which will serve as a fantastic tool that they can use to establish the mentality required to achieve long-term and short-term goals and teach younger athletes visualization, it's important to do it in a way that's age appropriate. Finally, future researchers can investigate into more aspects of how visualization can be detrimental to athletes, if such a thing exists.

Author Contributions

This work was carried out in collaboration among all authors. All authors read and approved the final manuscript.

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